Section 14.3: Partial Derivatives

Partial Derivatives

Partial Derivatives — Motivation and Definition

Partial Derivatives

$$f_x(x,y) = \lim_{h\to 0} \frac{f(x+h,\, y) - f(x,y)}{h}$$

$$f_y(x,y) = \lim_{k\to 0} \frac{f(x,\, y+k) - f(x,y)}{k}$$

These are just Calc I derivatives in disguise. The only new idea: **hold the other variable constant**.

Computing Partial Derivatives

You already know how to do this! Every derivative rule from Calc I (power, product, quotient, chain) still applies — you're just ignoring one variable.

Example 1: Polynomial Partials

Example 1

Find f_x and f_y for $f(x,y) = 3x^2y + xy^2 + y - 1$.

For f_x, treat y as a constant. For f_y, treat x as a constant.

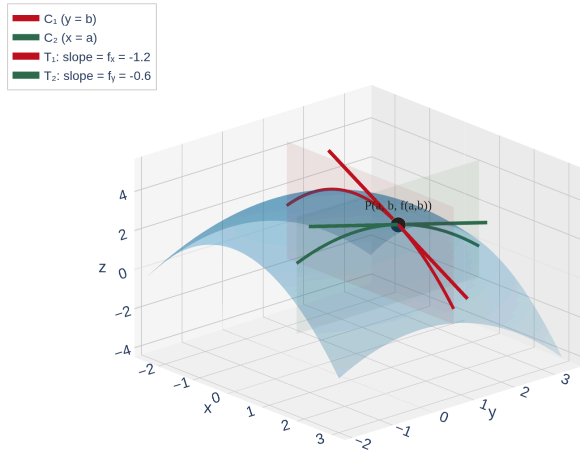
Your Turn: Compute Partials

Example 2

Find f_x and f_y for $f(x,y) = x^2 \ln y + y \sin x$.

For f_x, treat y as a constant — which terms involve x? For f_y, treat x as a constant.

Geometric Interpretation of Partial Derivatives



Interactive version

$f_x(a, b)$ = slope in the x**-direction**. $f_y(a, b)$ = slope in the y**-direction**. Each partial derivative captures only **one direction** of change.

Example 3: Different Rules for Different Variables

Example 3

Find both partial derivatives of $f(x, y) = x^y$.

This is interesting because x and y play **different roles** — one is the base, one is the exponent.

For f_x, treat y as constant (power rule). For f_y, treat x as constant (exponential rule).

Example 4: Rates of Change — Heat Index

Example 4

The heat index I is the perceived temperature as a function of actual temperature T and relative humidity H. Use the table to estimate $f_T(96, 70)$ and $f_H(96, 70)$, and interpret each.

Use the difference quotient with nearby table values. Hold one variable constant while varying the other.

Higher-Order Partial Derivatives

Interpretation:

- f_{xx} = concavity in the x-direction (is the surface curving up or down as you walk in x?)
- f_{yy} = concavity in the y-direction
- f_{xy} = how the x-slope changes as you move in y (a **mixed** partial)

Clairaut's Theorem

Theorem: Clairaut's Theorem

Suppose f is defined on a disk D containing the point (a, b).
If f_{xy} and f_{yx} are both **continuous** on D, then:

$$f_{xy}(a, b) = f_{yx}(a, b)$$

For any function built from standard Calc I functions, the **order of differentiation doesn't matter**. Differentiate in whichever order is easiest!

The Clairaut Puzzle

Example 5

You are told that there is a function f whose partial derivatives are:

$$f_x(x,y) = x + 4y \qquad \text{and} \qquad f_y(x,y) = 3x - y$$

Should you believe it?

Check whether Clairaut's theorem is satisfied.

Example 6: Laplace's Equation

Example 6

Verify that $u = \ln\sqrt{x^2+y^2}$ satisfies **Laplace's equation**:

$$u_{xx} + u_{yy} = 0$$

Laplace's equation appears throughout physics — it describes steady-state heat flow, electrostatics, and fluid flow.

Simplify u first, then find u_x, u_xx, u_y, u_yy, and add the second derivatives.

Example 7: Implicit Partial Differentiation

Example 7

If $yz = \ln(x+z)$, find $\dfrac{\partial z}{\partial x}$.

When z is defined **implicitly** as a function of x and y, we differentiate both sides with respect to x, treating y as a constant and z as a function of x.

Differentiate both sides with respect to x. Remember z depends on x, so use the chain rule.

Homework

Section 14.3: 1, 2, 4, 5, 6, 9-36 odd, 39, 41, 45, 51, 63, 64, 80, 84a, 85